

Colin Versteeg

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PROFESSIONAL EXPERIENCE

Meta

Ads Core ML
Software Engineer (E5)

Sept. 2023 -
Seattle, WA

- Implemented and productionized custom Triton GPU kernels for the two largest Meta ads models, driving 6% model QPS gains
- Debugged floating point accuracy issues caused by AOTInductor/TorchInductor lowering, GPU memory leaks and training/inference time inconsistencies with custom ops
- Built benchmarks and investigated kernel level performance, using NVIDIA Nsight Compute, PyTorch Profiler and Strobilight
- Led a team of 3 engineers to deliver scalability improvements to C++ CPU implementations of core preprocessing transforms for calibration models, saving 1 MW of serving power

Twitter

Home ML Systems
Machine Learning Engineer II

July 2020 - Aug. 2023
Seattle, WA

- Migrated the backend infrastructure of bookmarked Tweets to share a cache and storage with favorites, enabling it to scale from 60 queries per second to 1.5 million queries per second and launch bookmark counts on every tweet
- Led a team of 4 engineers to migrate the Home team's batch aggregation framework from Scalding to Apache Beam on Google Dataflow, improving performance by 3x and reducing cost by \$60K a month
- Launched the ranked home timeline in Twitter's third datacenter, serving over 30 million daily requests

Microsoft

AI Inferencing and Infrastructure
Software Engineer II

Nov. 2017 - July 2020
Redmond, WA

- Launched Azure Machine Learning Hardware Accelerated Models service, based on research from Microsoft Research, for acceleration of Neural Networks on Field Programmable Gate Arrays (FPGAs), delivering execution 50 times faster than CPUs
- Developed high performance C++ and C# gRPC servers to operationalize Tensorflow and ONNX graphs into a high performance DNN inference service on Kubernetes with sub-5ms overhead
- Built examples and documentation for how to deploy models on Azure Machine Learning and trigger them with asynchronous Azure Functions and presented them at internal conferences and Ignite 2019

Machine Learning Server
Software Engineer

Aug. 2016 - Nov. 2017

- Prototyped initial support for Python in Microsoft R Server and Microsoft SQL Server as part of a 4-person virtual team, leading to the launch of Microsoft Machine Learning Server and SQL Server Machine Learning Services
- Designed and implemented support for Python model operationalization in Microsoft Machine Learning Server

Internship Experience

2013 - 2015

- **Booz Allen Hamilton:** Developed prototype streaming data pipeline for malware analysis using Apache Storm, Elasticsearch, Java and Python.
- **FDA:** Created public-domain software for evaluating ROC curves for CT algorithm assessment studies using Python, NumPy and MATLAB; awarded ORISE fellowship.

EDUCATION

University of Maryland
B.S. Computer Engineering

May 2016
College Park, MD

SKILLS

Programming Languages: C++, Python, C#, Scala, Java, R, F#, Go

Platforms/Technologies: PyTorch, Triton, TensorFlow, Dataflow, .NET Core, Kubernetes, Azure, GCP